Md. Jubayer Hossain

Bioinformatics & Computational Biology · Prospective Ph.D. Student

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EDUCATION

Master of Science in Microbiology, Jagannath University, Dhaka, Bangladesh 2019 – 2020

Bachelor of Science in Microbiology, Jagannath University, Dhaka, Bangladesh 2016 – 2019

CERTIFICATIONS & HACKATHONS

scverse × Cell Painting Hackathon, scverse Consortium 2026

Interdisciplinary Computational Biology, BRAC University 2026

AI for Public Health, Child Health Research Foundation (CHRF) 2023

AI in Public Health, DeepLearning.AI 2022

TECHNICAL SKILLS

Programming: Python, R, Julia, SQL, Bash, JavaScript
Bioinformatics: RNA-seq (Salmon, DESeq2), scRNA-seq (Seurat, Scanpy, scvi-tools), spatial (Squidpy), multi-omics (Muon)
Workflows & Reproducibility: Nextflow (DSL2), nf-core, Docker, Singularity, Git, GitHub
ML/DL in Biology: PyTorch, scikit-learn, GNNs, RDKit, DeepChem
HPC & Cloud: SLURM, Nextflow SLURM executor; AWS S3, Google Cloud Storage

RESEARCH EXPERIENCE

Founder & Executive Director, CHIRAL Bangladesh (Institute for Volunteering Research) Jun 2020 – Present

Visiting Scholar, Dept. of Public Health, Daffodil International University Sep 2025 – Present

Research Collaborator, Kyung Hee University Jul 2026 – Present

Research Collaborator, Lamar University Jul 2026 – Present

Research Collaborator, One Health Lab, Jahangirnagar University Jul 2026 – Present

Research Intern, International Center for Diarrhoeal Disease Research, Bangladesh Feb 2022 – Jun 2022

Data Enumerator, School of Allied Health, Curtin University Oct 2022 – Nov 2022

PROFESSIONAL EXPERIENCE

Founder & CEO, DeepBio Limited Jan 2025 – Present

Faculty (Bioinformatics), cBLAST, University of Dhaka Aug 2023 – Present

Program Lead, GSA Bioinformatics Internship Jul 2025 – Present

Curriculum Developer and Instructor, School of Bioinformatics Jan 2024 – Present

Teaching Assistant, Department of Microbiology, Jagannath University Oct 2022 – Dec 2022

SELECTED PUBLICATIONS

Underline indicates mentee; * corresponding

1. Islam, M. W., Rahman, M. M., Naznin, H., Hossain, M. S., Akter, T., Shatabde, Z. A., & ***Hossain, M. J.** (2026) "Integrative Bioinformatics Analysis Reveals COL13A1 and COL23A1 as Potential Diagnostic and Prognostic Biomarkers in Thyroid Cancer" *Health Science Reports*. DOI: 10.1002/hsr2.72617
2. Fariha, F. T. J., Fuad, M., Saha, C. S., Hossen, S., & ***Hossain, M. J.** (2025) "Comprehensive bioinformatics analysis reveals prognostic significance and immunological roles of WNT gene family in breast cancer" *Scientific Reports*. DOI: 10.1038/s41598-025-13315-6
3. Ahmed, M. Z., Billah, M. M., Ferdous, J., & ***Hossain, M. J.** (2025) "Pan-cancer analysis reveals immunological and prognostic significance of CCT5 in human tumors" *Scientific Reports*. DOI: 10.1038/s41598-025-88339-z
4. Bari, S. M., Fuad, M., **Hossain, M. J.**, Ahammad, I., Begum, M., & Helal, M. M. U. (2025) "A meta-analysis of public RNA-Seq data identifies conserved stress responses in rainbow trout" *BMC Genomics*. DOI: 10.1186/s12864-025-12127-2
5. Kundu, P., Shahariar, M., Munny, F. A., Riya, A. T., Lota, H. B., Ahmed, M. Z., Meme, M. A., Kona, U. M. P., Mahmuda, F., Chowdhury, S. S., Sraboni, S. A., & ***Hossain, M. J.** (2026) "Computational identification of Taxus baccata-derived

phytochemicals targeting EZH2 to overcome therapeutic resistance in melanoma" *Scientific Reports* (Under Review). DOI: 10.21203/rs.3.rs-8494315/v1

6. Islam, M. W., Fariha, F. T. J., Ahmed, M. Z., Ferdous, J., Lota, H. B., Hossain, M. S., Kundu, P., Shahjahan, M., & *__Hossain, M. J.__ (2025) "Bioinformatics-driven multi-omics profiling of CDK1 and CDK6 identifies prognostic and therapeutic roles in breast cancer" *Health Science Reports* (Under Review).
7. Billah, M. M., Mabsurah, K., Ahammad, K., Yeana, I. J., Sumaiya, M., Islam, T., Bhattacharjee, A., Ferdous, J., & *__Hossain, M. J.__ (2025) "Multi-omics pan-cancer analysis reveals an immunological role and prognostic potential of WDR76" *BMC Genomics* (Under Review).
8. **Hossain, M. J.** (2026) "metaXpress: An End-to-End Bioconductor Framework for Quality Control, Harmonization, and Multi-Cohort Bulk RNA-seq Meta-Analysis" *bioRxiv*. GitHub
9. **Hossain, M. J.** (2026) "Graph-Based and Graph-Free Models for Spatial Niche Prediction in Glioblastoma" *bioRxiv*. GitHub
10. **Hossain, M. J.** (2026) "Translating Perturb-seq Screens Across Cell Lines" *bioRxiv*. GitHub
11. **Hossain, M. J.**, & Dev, P.C. (2026) "Evaluating the false-positive rate of single-cell copy-number and malignant-cell classifiers" *bioRxiv*. GitHub
12. **Hossain, M. J.**, Dev, P. C., Nawar, T., Arony, T. A., Hoque, M., Chowdhury, D., & Rahman, M. A. (2026) "A donor-level map of immune sex differences" *bioRxiv*. GitHub
13. **Hossain, M. J.**, Iqbal, M. (2026) "EWSAtlas: A Harmonized Single-Cell Transcriptomic Atlas of Ewing Sarcoma" *Working paper*.
14. **Hossain, M. J.**, Iqbal, M. (2026) "Unified Single-Cell Atlas of Diffuse Midline Glioma (DMG)" *Working paper*.

SELECTED PRESENTATIONS

1. Rahman, M. A., Shahjahan, M. and *__Hossain, M. J.__ EBV-antigen-specific T cell clonotype fate in nasopharyngeal carcinoma: A single-cell TCR analysis. Dhaka University Research Society (DURS), December, 2026.
2. Chowdhury, D., Shahjahan, M. and *__Hossain, M. J.__ Reassessing the chemotherapy–TIGIT association in pancreatic cancer through cross-cohort validation. Dhaka University Research Society (DURS), December, 2026.
3. Hoque, M., Shahjahan, M. and *__Hossain, M. J.__ Chemotherapy-induced tumor cell state dynamics in osteosarcoma: NMF-based decomposition and independent bulk validation. Dhaka University Research Society (DURS), December, 2026.
4. Arony, T. A., Shahjahan, M. and *__Hossain, M. J.__ Spatial mapping of diffuse midline glioma cell states reveals conserved territorial architecture and a mesenchymal-like interface for tumor–microenvironment crosstalk. Dhaka University Research Society (DURS), December, 2026.
5. Nawar, T., Shahjahan, M., *__Hossain, M. J.__ Cross-Cohort Meta-Analysis of Acral Melanoma Immune Cell Quantity vs. Function With Three-Way Decomposition. Dhaka University Research Society (DURS), December, 2026.
6. Islam Abhi, S. N., Meme, M. A., Shahariar, M., Barsha, L., *__Hossain, M. J.__ Single-cell transcriptomic and network pharmacology insights into the therapeutic potential of Withania somnifera-derived phytochemicals against cisplatin-resistant ovarian cancer. BEICTCON2026, IEEE, August, 2026.
7. Islam, M. M., *__Hossain, M. J.__ Fezf2-Mediated Cortical Development: Multi-Omics Single-Cell Analysis. Network of Young Biotechnologists Bangladesh, April, 2026.
8. Ahmad, F., Fuad, M., Shahjahan, M. and *__Hossain, M. J.__ Identification of Subtype-Specific Prognostic Biomarkers in Esophageal Carcinoma through Integrated Bioinformatics Analysis. Bangladesh Society of Microbiologists, December, 2025.

AWARDS & FELLOWSHIPS

Open Bioinformatics Foundation Event Fellowship, *Open Bioinformatics Foundation* 2026
Travel Grant for scverse x cellpainting Hackathon, *scverse* 2026
Scholarship for AI in Public Health Workshop, *Bill and Melinda Gates Foundation* 2023
Dutch-Bangla Bank (DBBL) Merit Scholarship, *Dutch-Bangla Bank* 2013

REFERENCES

Dr. Syeda Tasneem Towhid, Associate Professor, Dept. of Microbiology, Jagannath University towhidst@mib.jnu.ac.bd
Dr. Iqbal Mahmud, Assistant Professor, Lamar University, TX, USA imahmud1@lamar.edu
Dr. Zeba Islam Seraj, Director, cBLAST, University of Dhaka zebai@du.ac.bd
Dr. Yeon Ju Kim, Professor, Convergent Biotechnology, Kyung Hee University, South Korea yeonjukim@khu.ac.kr